Customer Dataset — Data Cleaning Report

Prepared from data-cleaning.ipynb | E-commerce Customer Dataset

Dina Ebrahim Nadjari
github.com/dinanadjari

This report summarizes the data-cleaning and exploratory analysis workflow applied to an e-commerce customer dataset (customer demographics, membership tier, spending, and satisfaction behaviour). The raw file was inspected, corrected, and standardized before being exported as a clean, analysis-ready dataset.

Dataset Snapshot

17 Columns / Features

54 Clean Records Retained

6 Issue Categories Fixed

1 Excel Export File

Original Column Inventory

The source workbook (First Dataset.xlsx) contained 17 fields describing each customer: identity & demographics (customer_id, first_name, gender, age, city, province), account & engagement (signup_date, membership_tier, last_purchase_days), transaction behaviour (purchase_count, avg_order_value, total_spending, returned_items, discount_used, payment_method, device), and feedback (satisfaction_score).

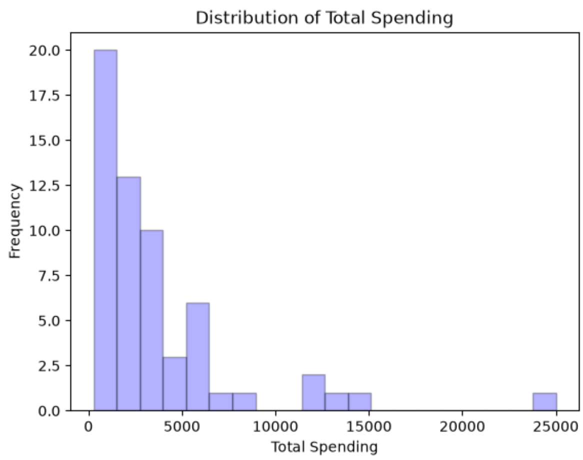
Data Quality Issues Identified

Duplicate row	One fully duplicated customer record found via row-level duplicate check.	all columns
Impossible age	Maximum age value of 145 detected — biologically implausible.	age
Missing values	Nulls present in age and total_spending after outlier removal.	age, total_spending
Incorrect gender labels	Gender values inconsistent with customer first names.	gender
Logical contradiction	6 records where returned_items exceeded purchase_count.	purchase_count, returned_items
Invalid transaction	Record with zero total_spending but non-zero avg_order_value.	total_spending, avg_order_value
Inconsistent encoding	discount_used stored as Yes/No text instead of a binary flag.	discount_used

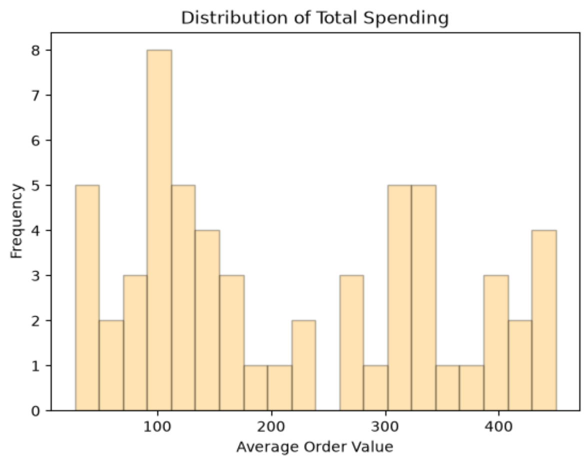
Cleaning Steps Applied

#	Step	Action Taken
1	Remove duplicates	Dropped the exact duplicate customer row using drop_duplicates().
2	Fix age outliers	Ages outside a realistic 10–90 range were set to NaN, then imputed with the column median.
3	Impute missing values	Remaining nulls in age and total_spending filled with their respective medians.
4	Correct gender labels	Rebuilt the gender column from a verified first-name-to-gender mapping.
5	Flag logical contradictions	Records with returned_items > purchase_count had both fields set to NaN pending a reliable source.
6	Remove invalid record	Dropped the customer with zero spending yet a non-zero average order value (customer_id 1024).
7	Standardize data types	Converted signup_date to proper datetime; encoded discount_used as 0/1.

Spending Distributions (Post-Cleaning)



Total spending is right-skewed, driven by a small group of high-value VIP customers.



Average order value is broadly spread, with no single dominant price tier.

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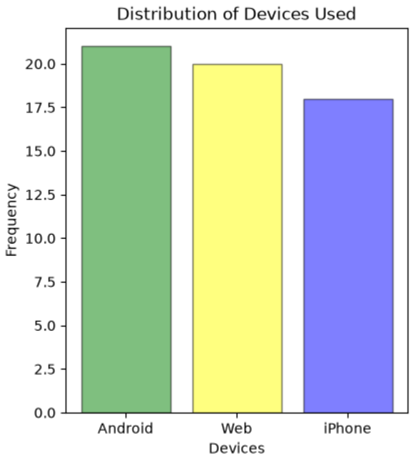
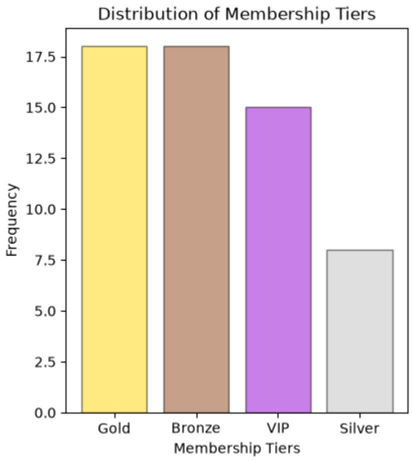
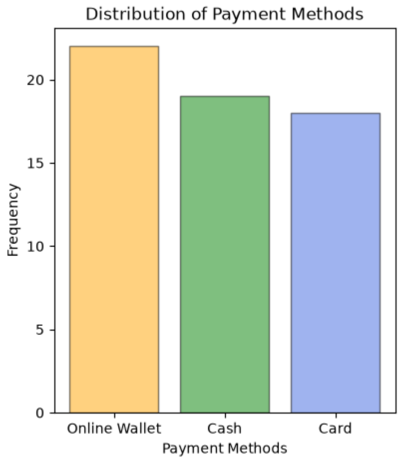
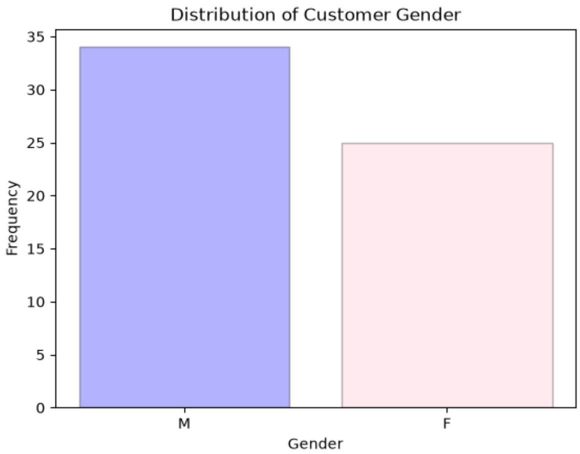
Customer Profile Distributions



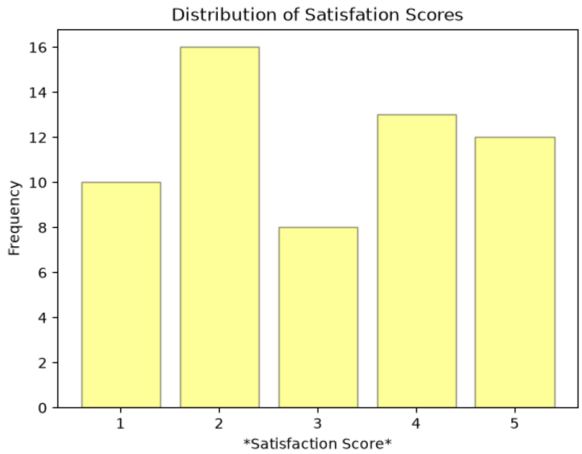




Device usage, payment method, and membership tier are all fairly balanced, with Gold and Bronze the leading membership tiers.



Customer base skews female (35) vs. male (25) after gender correction.



Satisfaction scores cluster around 2, suggesting room to improve customer experience.

Summary & Recommendations

After removing duplicates, correcting impossible or inconsistent values, and standardizing data types, the dataset was reduced to 54 reliable customer records and exported to **cleaned_dataset.xlsx**. The data is now consistent enough for downstream analysis such as customer segmentation, churn risk modeling, or satisfaction-driver analysis.

Investigate low satisfaction scores	Score 2 is the most frequent rating — worth root-causing.
Source a gender-verification process	Reduces reliance on name-based inference for future records.
Add data validation at entry point	Prevents impossible ages and negative logical values upstream.